Supplementary Information

RAFT-Ports: An Application for Risk Assessment for Tsunamis in Ports Demonstrated at Tanjung Gerem Port, Indonesia

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Table S1. Comparison of existing tsunami hazard, vulnerability, and risk assessment tools with RAFT-Ports. The installers that are inaccessible at the time of review are marked with *.

Tool	Geographic Applicability	Analysis Approach	Tsunami-Specific	Port-Specific	Structural Damage	Loss Ratio	Recovery
RAFT-Ports	**Global (user-defined)**	**User-defined analysis using custom inundation & fragility data**	**Yes**	**Yes**	**Yes**	**Yes**	**Yes**
EPOS	Selected regions	Primarily pre-generated hazard/risk products	Yes	No	Limited	No	No
ETRiS	Global	User-defined analysis	Yes	No	Yes	No	No
HAZUS*	Primarily USA	User-defined analysis within HAZUS framework	Yes	No	Yes	Yes	Limited
InaRisk	Indonesia	Primarily pre-generated hazard and risk products	Yes	No	Yes	Yes	No
InaSAFE	Global (user data required)	User-defined analysis using custom hazard and exposure datasets	Multi-hazard	No	Yes	Limited	No
INSPIRE/rIMES*	Selected regions	Primarily pre-generated inundation and risk products	Yes	No	Yes	No	No
InVEST	Global	User-defined analysis	No	No	No	No	No
TsuCAT	Global	User-defined tsunami scenario exploration	Yes	No	No	No	No



Figure S1. The 100 slip models used for the Mw 8.5 event.



Figure S2. The 100 slip models used for the Mw 8.7 event.

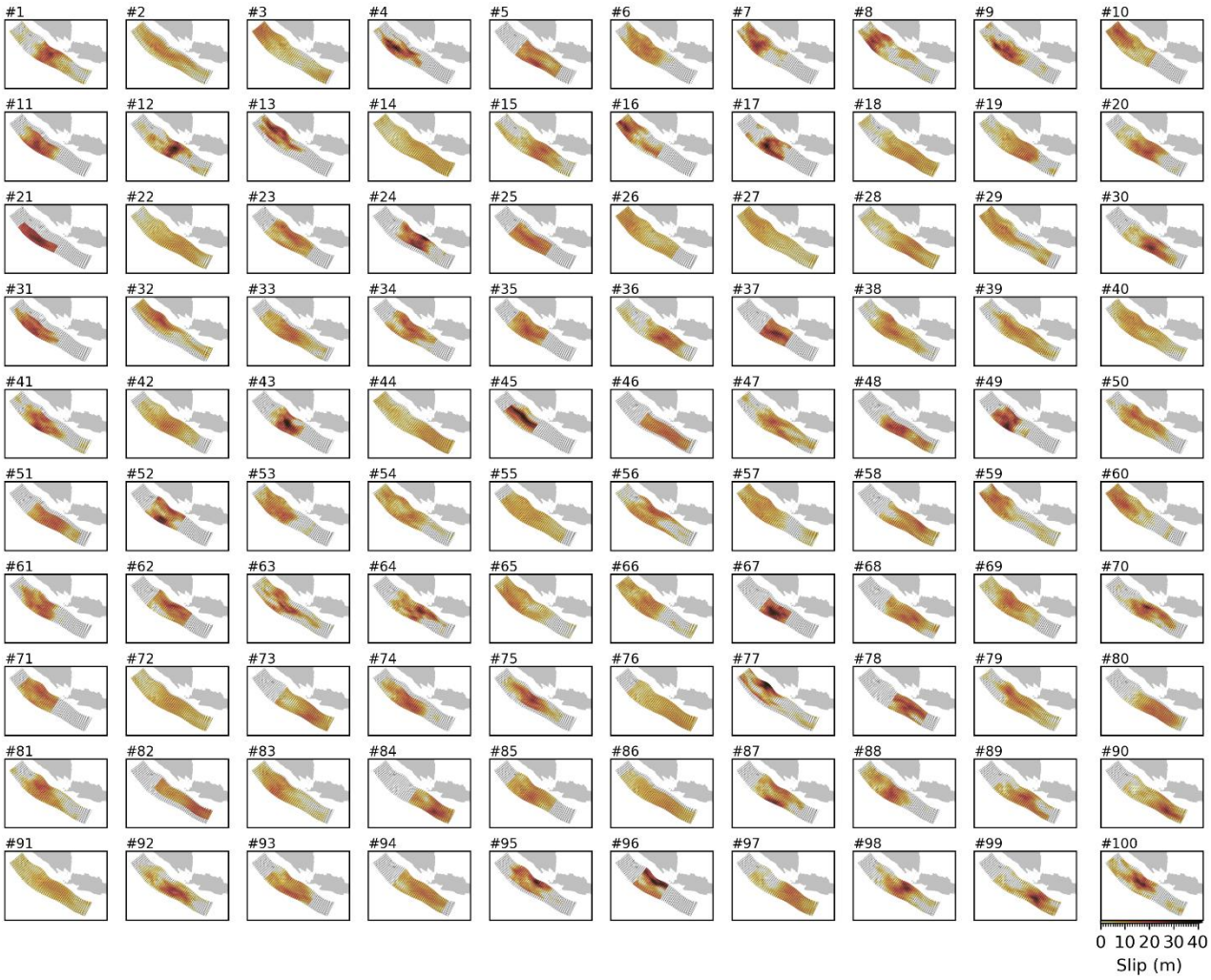


Figure S3. The 100 slip models used for the Mw 8.9 event.

Table S2. Summary of slip values and fault-plane area for the three earthquake scenarios. 'SD' stands for standard deviation. Corresponding maps are presented in Figure S2.

Mw	Slip (m)		Area (km²)	
	Mean	SD	Mean	SD
8.5	4.84	1.62	47,385	16,284
8.7	7.28	2.73	63,651	23,255
8.9	9.87	3.35	90,440	26,084

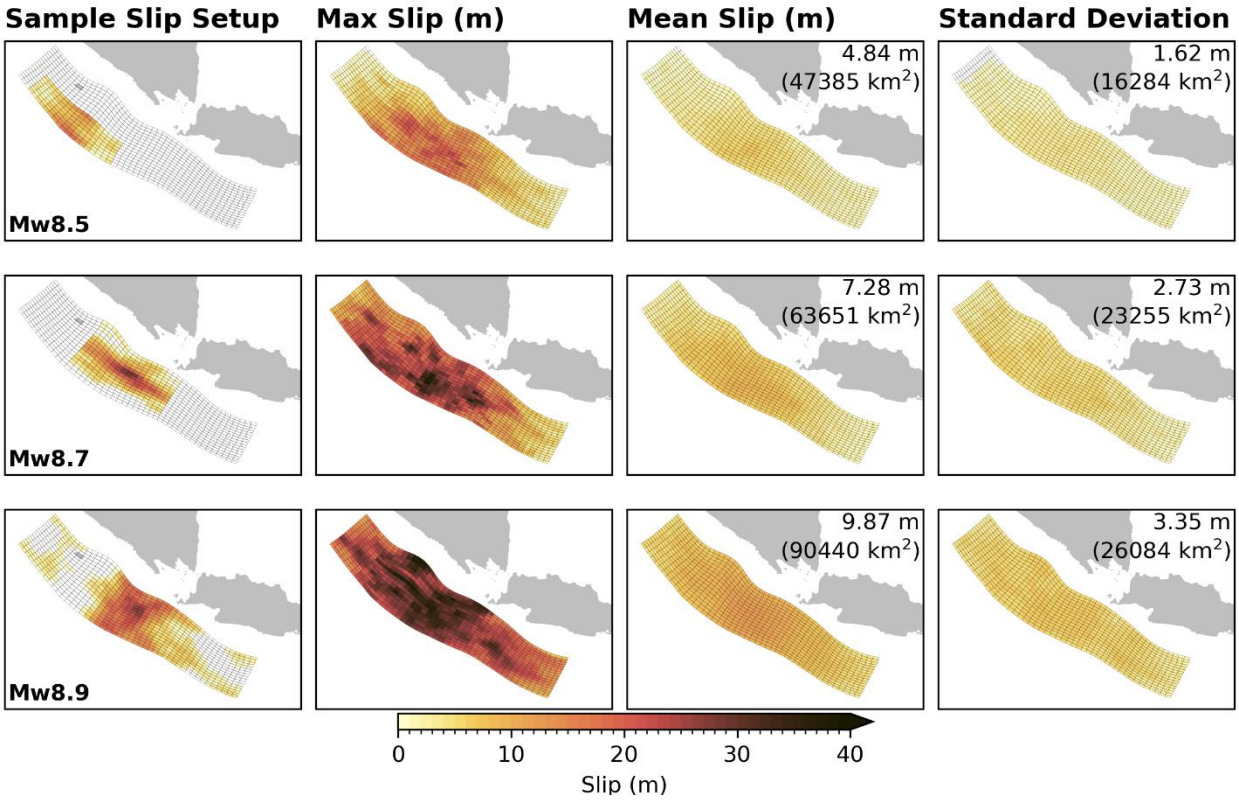


Figure S4. Summary of slip values and fault plane area for the three earthquake magnitudes.

Table S3. Grid spacing and extent of the nested grids used in numerical modelling.

Layer	Grid Spacing		Model Extent	
	Arc degrees	**Metres**	**Longitude**	**Latitude**
1	0.01158	1280	[88.595, 119.408]	[-19.140, 2.977]
2	0.00287	320	[104.496, 106.248]	[-6.496, -5.416]
3	0.00072	80	[105.299, 106.083]	[-6.254, -5.736]
4	0.00018	20	[105.825, 106.010]	[-6.019, -5.927]
5	0.00005	5	[105.957, 106.006]	[-6.000, -5.953]

Figure S5. Tsunami wave records at Gauge 1, located at the 100 m depth contour at the southwestern entrance of the channel between Sangiang Island and Java (towards Tanjung Gerem Port, Figure 2e). Panels (a)–(c) show the 100 stochastic scenarios generated for M_w 8.5, M_w 8.7, and M_w 8.9, respectively. The highlighted blue curve represents the worst-case scenario for each magnitude, selected based on the maximum wave height recorded at the gauge.

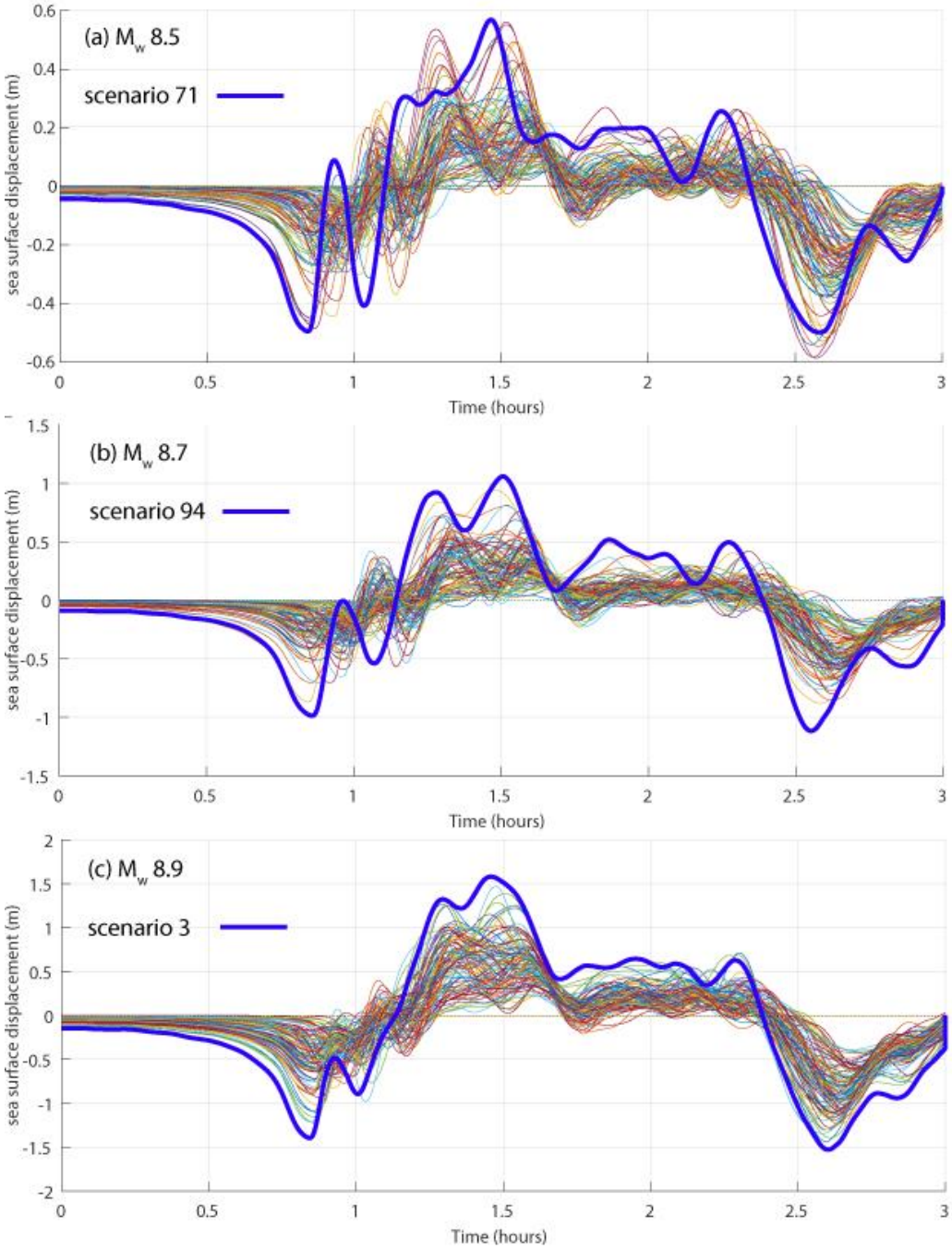


Table S4. Chemical Industry: Probability-of-exceedance values for each depth range, derived from the average fragility curves used in the Structural Damage State analysis (Chua et al., 2021), and expected loss-ratio values derived from the average vulnerability functions used in the Loss Ratio analysis (Miki et al., 2025). Note: Although these values are the default settings in RAFT--Ports, they can be modified by the user.

Depth (m)		Structural Damage State				Loss Ratio
Range	Reference	Minor	Moderate	Major	Destroyed	
(0 .0 – 0.5]	0.5	0.45	0.0047	0.0012	0.0004	0.3233
(0.5 – 1.0]	1.0	0.8988	0.0488	0.0129	0.0042	0.4818
(1.0 – 1.5]	1.5	0.9731	0.171	0.0501	0.0171	0.5623
(1.5 – 2.0]	2.0	0.9898	0.3563	0.1243	0.0448	0.6535
(2.0 – 2.5]	2.5	0.9952	0.5446	0.2346	0.0918	0.7446
(2.5 – 3.0]	3.0	0.9974	0.6918	0.3652	0.1589	0.8193
(3.0 – 3.5]	3.5	0.9985	0.7931	0.4957	0.2436	0.8729
(3.5 – 4.0]	4.0	0.9991	0.859	0.609	0.3381	0.9095
(4.0 – 4.5]	4.5	0.9994	0.9016	0.6999	0.4334	0.9344
(4.5 – 5.0]	5.0	0.9996	0.9293	0.7706	0.5236	0.9516
(5.0 – 5.5]	5.5	0.9997	0.948	0.8226	0.6039	0.9634
(5.5 – 6.0]	6.0	0.9998	0.9607	0.8625	0.6727	0.972
(6.0 – 6.5]	6.5	0.9998	0.9699	0.892	0.7305	0.9781
(6.5 – 7.0]	7.0	0.9999	0.9765	0.9139	0.7774	0.9827
(7.0 – 7.5]	7.5	0.9999	0.9814	0.9308	0.816	0.9861
(7.5 – 8.0]	8.0	0.9999	0.9765	0.9438	0.8475	0.9887
(8.0 – 8.5]	8.5	0.9999	0.9878	0.9539	0.8727	0.9907
(8.5 – 9.0]	9.0	0.9999	0.9899	0.9618	0.8928	0.9923
(9.0 – 9.5]	9.5	1	0.9916	0.968	0.9092	0.9935
(9.5 – 10]	10.0	1	0.9899	0.9731	0.9227	0.9945

Table S5. Petrochemical Industry: Probability-of-exceedance values for each depth range, derived from the average fragility curves used in the Structural Damage State analysis (Chua et al., 2021), and expected loss-ratio values derived from the average vulnerability functions used in the Loss Ratio analysis (Miki et al., 2025). Note: Although these values are the default settings in RAFT--Ports, they can be modified by the user.

Depth (m)		Structural Damage State				Loss Ratio
Range	Reference	Minor	Moderate	Major	Destroyed	
(0 .0 – 0.5]	0.5	0.6229	0.0005	0.0001	0.0000	0.3462
(0.5 – 1.0]	1.0	0.9654	0.0086	0.0024	0.0006	0.4851
(1.0 – 1.5]	1.5	0.9933	0.0433	0.0122	0.0031	0.5331
(1.5 – 2.0]	2.0	0.9979	0.1276	0.0381	0.0100	0.6098
(2.0 – 2.5]	2.5	0.9992	0.2660	0.0895	0.0244	0.7091
(2.5 – 3.0]	3.0	0.9996	0.4328	0.1718	0.0499	0.8007
(3.0 – 3.5]	3.5	0.9998	0.5888	0.2813	0.0901	0.8686
(3.5 – 4.0]	4.0	0.9999	0.7118	0.4021	0.1461	0.9136
(4.0 – 4.5]	4.5	0.9999	0.7994	0.5197	0.2165	0.9425
(4.5 – 5.0]	5.0	1.0000	0.8594	0.6252	0.2991	0.9605
(5.0 – 5.5]	5.5	1.0000	0.9002	0.7105	0.3867	0.9724
(5.5 – 6.0]	6.0	1.0000	0.9279	0.7783	0.4739	0.9803
(6.0 – 6.5]	6.5	1.0000	0.9469	0.8297	0.5556	0.9855
(6.5 – 7.0]	7.0	1.0000	0.9601	0.8688	0.6286	0.9891
(7.0 – 7.5]	7.5	1.0000	0.9697	0.8975	0.6919	0.9917
(7.5 – 8.0]	8.0	1.0000	0.9765	0.9192	0.7459	0.9935
(8.0 – 8.5]	8.5	1.0000	0.9816	0.9359	0.7897	0.9949
(8.5 – 9.0]	9.0	1.0000	0.9854	0.9485	0.8260	0.9960
(9.0 – 9.5]	9.5	1.0000	0.9883	0.9583	0.8553	0.9967
(9.5 – 10]	10.0	1.0000	0.9905	0.9660	0.8794	0.9973

Table S6. Probability-of-exceedance values for the potential recovery of production capacity derived from the average line values of the fragility curves developed by Suppasri et al. (2022). Note: Although these values are the default settings in RAFT-Ports, they can be edited by the user.

Months		Inundation Depth at Buildings		
Range	Reference	0 m	3 m	≥3 m
(0 .0 – 0.5]	0.5	0.015	0	0
(0.5 – 1.0]	1.0	0.070	0.005	0.005
(1.0 – 1.5]	1.5	0.215	0.020	0.015
(1.5 – 2.0]	2.0	0.460	0.045	0.030
(2.0 – 2.5]	2.5	0.720	0.090	0.060
(2.5 – 3.0]	3.0	0.900	0.160	0.110
(3.0 – 3.5]	3.5	0.980	0.255	0.180
(3.5 – 4.0]	4.0	1.000	0.380	0.275
(4.0 – 4.5]	4.5	1.000	0.515	0.385
(4.5 – 5.0]	5.0	1.000	0.650	0.505
(5.0 – 5.5]	5.5	1.000	0.765	0.625
(5.5 – 6.0]	6.0	1.000	0.855	0.735
(6.0 – 6.5]	6.5	1.000	0.920	0.825
(6.5 – 7.0]	7.0	1.000	0.960	0.890
(7.0 – 7.5]	7.5	1.000	0.980	0.940
(7.5 – 8.0]	8.0	1.000	0.995	0.965
(8.0 – 8.5]	8.5	1.000	1.000	0.985
(8.5 – 9.0]	9.0	1.000	1.000	0.995
(9.0 – 9.5]	9.5	1.000	1.000	1.000
(9.5 – 10]	10.0	1.000	1.000	1.000

Table S7. Brief description of the structural damage states defined by Chua et al. (2021) and the corresponding labels used in our risk analysis.

Label	Damage State (DS)	Description	Serviceability
Minor	DS 1	Inundated but the non-structural components and roof are intact.	Functional but requires interior fixing
Moderate	DS 2	Sustained damage on non-structural components and/or roofs.	Operational after repairs
Major	DS 3	Sustained damage on structural components, or buckling/folding of rackings; building collapse is possible	Not repairable
Destroyed	DS 4	Total structural failure.	Not operational

Table S8. Comparison of 'DepthGroup' assignment mismatches between RAFT-Ports and the independently replicated QGIS-Excel workflow for different inundation-depth extraction and raster-resampling settings. Mismatch rates remained low (≤2.3% of 430 structures), indicating only minor differences attributable to software-specific raster-processing algorithms rather than methodological inconsistencies. Figures S6 and S7 further illustrate how the number of mismatches varies with the selected inundation-depth extraction method and raster-resampling approach. The raw data is in 'Validation_QGIS_vs_RAFT-Ports.xls' which can be accessed at github.com/raquelpfelix/RAFT-Ports.

	Count of mismatch relative to the results from QGIS					
	Raw Count			% out of 430 structures		
Resolution of the Inundation File	Extracted Depth			Extracted Depth		
	Max	Mean	Min	Max	Mean	Min
Original Resolution	3	3	3	0.7	0.7	0.7
Cubic (2x)	4	4	9	0.9	0.9	2.1
Cubic (3x)	4	5	5	0.9	1.2	1.2
Cubic (4x)	6	5	10	1.4	1.2	2.3
Linear (2x)	3	2	7	0.7	0.5	1.6
Linear (3x)	1	3	3	0.2	0.7	0.7
Linear (4x)	1	4	6	0.2	0.9	1.4
Nearest (2x)	0	0	0	0.0	0.0	0.0
Nearest (3x)	1	3	0	0.2	0.7	0.0
Nearest (4x)	2	5	4	0.5	1.2	0.9

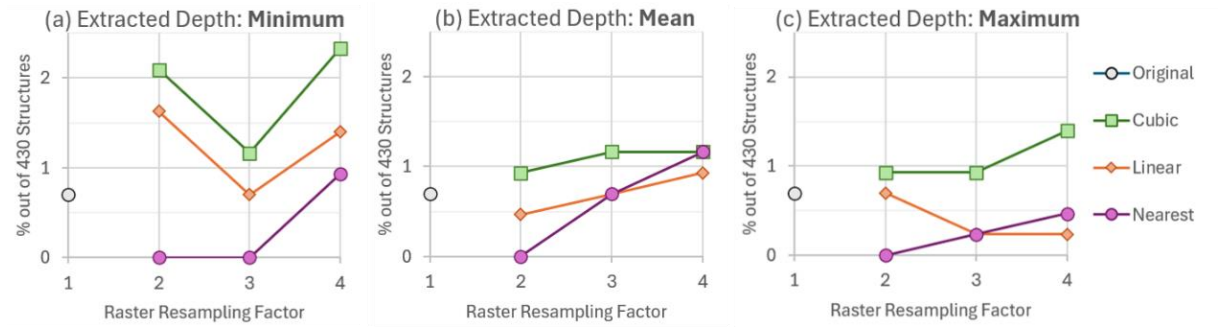


Figure S6. Comparison of 'DepthGroup' assignment mismatches between RAFT-Ports and the independently replicated QGIS-Excel workflow for different inundation-depth extraction methods. A resampling factor of 1 represents the original raster resolution (~5 m) used in the numerical modelling, while higher resampling factors correspond to progressively finer raster grid resolutions. Across all tested settings, mismatch rates remained low (<2.5% of 430 structures). In general, using the 'Nearest' resampling method, resulted in the fewest mismatches between the two workflows. These results indicate that 'DepthGroup' assignments are only minimally affected by the choice of inundation-depth extraction method, raster-resampling method, and raster resolution. The corresponding raw values used to generate this figure are provided in Tables S8.

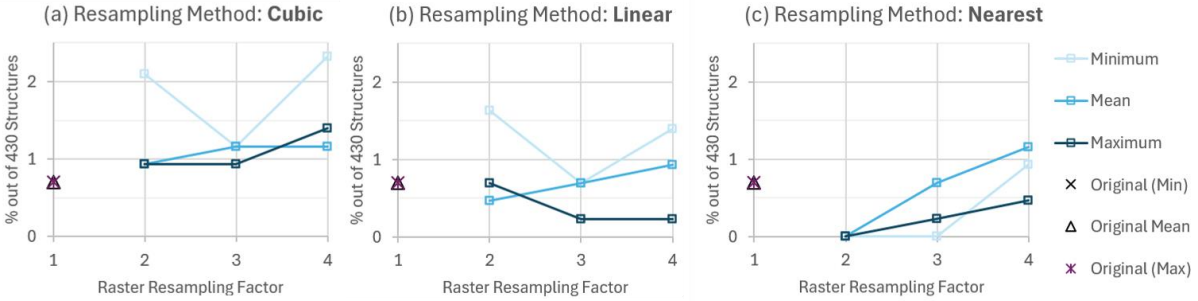


Figure S7. Comparison of 'DepthGroup' assignment mismatches between RAFT-Ports and the independently replicated QGIS-Excel workflow for different raster-resampling settings and inundation-depth extraction methods. A resampling factor of 1 corresponds to the original inundation raster resolution (~5 m), while higher factors represent progressively finer raster resolutions. Across all resampling methods and factors, mismatch rates remained low (<2.5% of 430 structures). In general, the maximum-depth extraction method resulted in the fewest mismatches for the resampled rasters, indicating that 'DepthGroup'assignments are only minimally affected by the choice of inundation-depth extraction method and raster-resampling settings. The corresponding raw values used to generate this figure are provided in Table S8.

Table S9. Raw counts (upper panel) and percentage counts (lower panel) used to generate Figures S8 to S10. The values show how changes in threshold value affect the number of structures reported by RAFT-Ports. The raw data is in 'Validation_QGIS_vs_RAFT-Ports.xls' which can be accessed at github.com/raquelpfelix/RAFT-Ports.

Raw Count

Threshold	Structural Damage States				Loss Ratio	Potential Recovery Period of Production Capacity			
	Minor	Moderate	Major	Destroyed		Start of Recovery*	# of structures	Full Recovery	# of structures
0.3	418	160	31	0	418	(1.5-2.0]	12	3.5-4.0	430
0.5	416	70	0	0	354	(2.0-2.5]	12	4.0-4.5	430
0.7	416	0	0	0	130	(2.0-2.5]	12	5.0-5.5	430
0.9	354	0	0	0	0	(2.5-3.0]	12	6.0-6.5	430

Percentage out of the 430 structures

Threshold	Structural Damage States				Loss Ratio	Potential Recovery Period of Production Capacity			
	Minor	Moderate	Major	Destroyed		Start of Recovery*	# of structures	Full Recovery	# of structures
0.3	97.2	37.2	7.2	0.0	97.21	(1.5-2.0]	2.79	3.5-4.0	100
0.5	96.7	16.3	0.0	0.0	82.33	(2.0-2.5]	2.79	4.0-4.5	100
0.7	96.7	0.0	0.0	0.0	30.23	(2.0-2.5]	2.79	5.0-5.5	100
0.9	82.3	0.0	0.0	0.0	0.00	(2.5-3.0]	2.79	6.0-6.5	100

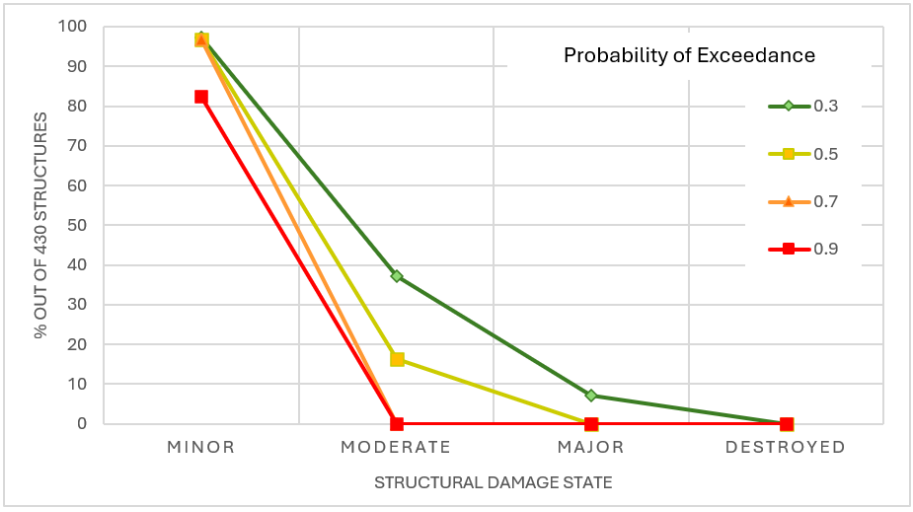


Figure S8. Differences in the reported percentage of structures resulting from changes in the probability-of-exceedance threshold for structural damage states. Increasing the threshold results in a decrease in the number of structures exceeding the selected criterion and therefore included in the reported counts. The corresponding raw values used to generate this figure are provided in Table S9.

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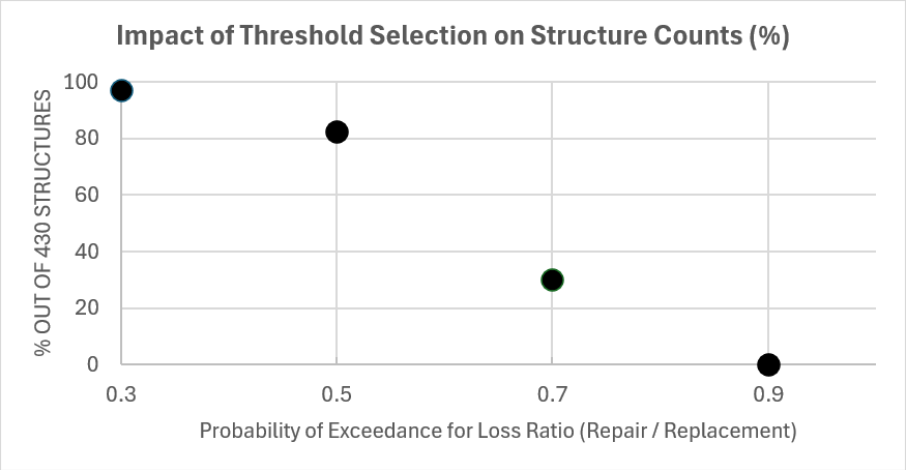


Figure S9. Differences in the reported percentage of structures resulting from changes in the threshold value for loss-ratio analysis. Increasing the threshold results in a decrease in the number of structures exceeding the selected criterion and therefore included in the reported counts. The corresponding raw values used to generate this figure are provided in Table S9.

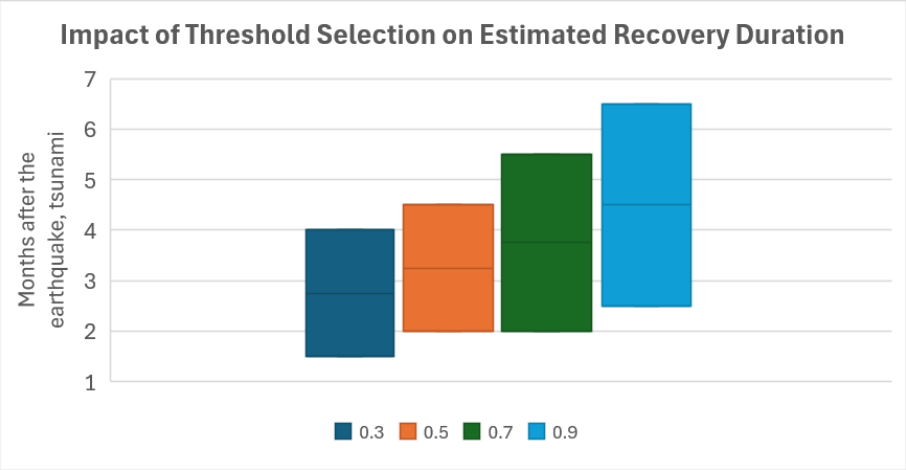


Figure S10. Differences in the reported estimated recovery duration of production capacity resulting from changes in the probability-of-exceedance threshold for recovery potential. Increasing the threshold (from 0.3 to 0.9) results in a later onset of potential recovery and a longer period before recovery probabilities exceed the selected threshold. The corresponding raw values used to generate this figure are provided in Table S9.

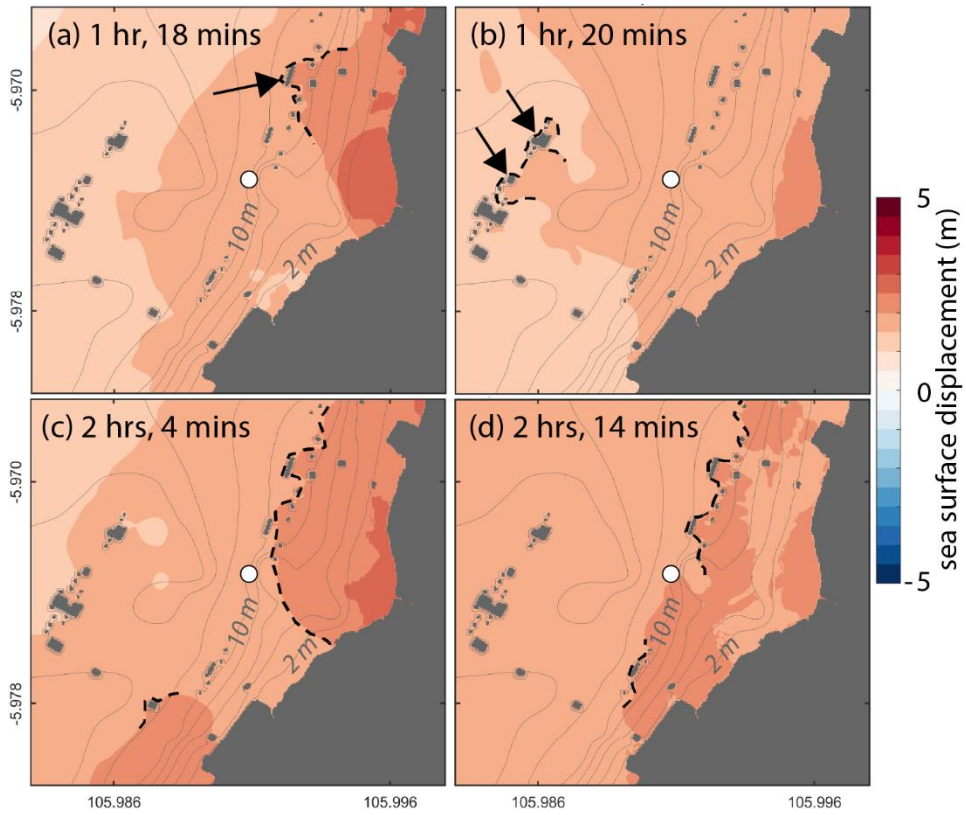


Figure S11. Close-up view of tsunami heights for selected time snapshots of the Mw 8.9 scenario, showing the influence of the jetty piles. Figures (c) and (d) also show that the piles, together with the convex coastline, cause wave entrapment and amplification. Wave refraction behind the piles (facing the main coast) is shown by dashed lines and arrows. White dot – virtual gauge station.

Table S10. Breakdown of the number of structures for the Structural Damage State (SDS), separated by industry type (Chemical and Petrochemical). The low counts reported in this table reflect the relatively high probability-of-exceedance threshold of 0.70 adopted in the analysis. As shown in Figure S8, reducing the threshold value results in a greater number of structures exceeding the selected threshold and, consequently, higher counts across the damage-state categories.

Scenario	Structural Damage State (%)							
	Chemical (n = 346)				Petrochemical (n = 84)			
	Minor	Moderate	Major	Destroyed	Minor	Moderate	Major	Destroyed
Mw 8.9	96	0	0	0	100	0	0	0
Mw 8.7	67	0	0	0	100	0	0	0
S9Mw 8.5	1	0	0	0	26	0	0	0

Table S11. Breakdown of the number of structures for the loss ratio, separated by industry type (Chemical and Petrochemical). The low counts reported in this table reflect the relatively high threshold value of 0.70 adopted in the analysis. As shown in Figure S9, reducing the threshold value results in a greater number of structures exceeding the selected threshold.

Scenario	Loss Ratio (%)	
	Chemical (n = 346)	Petrochemical (n = 84)
Mw 8.9	16	77
Mw 8.7	5	0
Mw 8.5	0	0